

Limits.

Sunday, June 9, 2024 7:30 AM

$$0 \times \infty = ? \quad , \quad \frac{0}{0} = ? \quad \frac{\infty}{\infty} = ?$$

$$\frac{0}{0}, \infty - \infty, \frac{\infty}{0}, 0 \times \infty, 1^\infty, 0^\infty, \infty^0 \quad (\text{undefined})$$

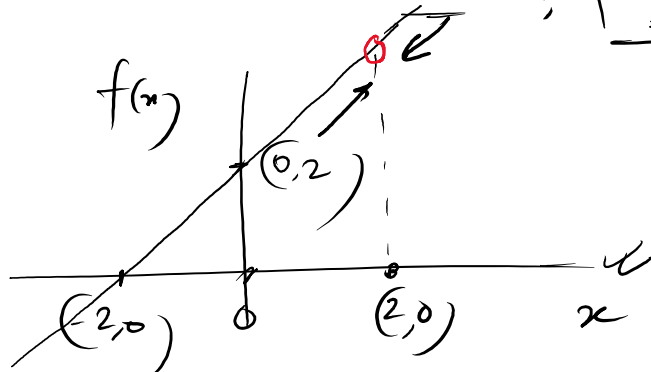
$$\frac{0}{0} = \text{undefined} \quad , \quad \frac{0.000000 - .1}{0.0000 - .1} = 1$$

$$f(x) = \frac{x^2 - 4}{x - 2} = \frac{(x+2)(x-2)}{(x-2)} \quad x \neq 2$$

domain $\mathbb{R} - \{2\}$

$$f(x) = x+2 \quad , \quad x \neq 2$$

undefined, $\boxed{x=2}$



$$f(2) = \text{undefined!}$$

$$\lim_{x \rightarrow 2} f(x) = 4$$

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e, \quad y = 1/x$$

$\uparrow$

$$y \rightarrow \infty$$

$$\lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right)^y \leftrightarrow \lim_{x \rightarrow 0} \left(1 + \frac{1}{x}\right)^x = e$$

$$Z = (1+x)^{1/x}, \quad x \rightarrow 0$$

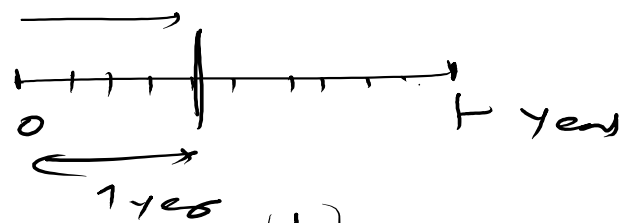
$$\ln(Z) = \left[\frac{1}{x} \ln(1+x) \right] = 1$$

$$Z = e$$

$$\lim_{m \rightarrow \infty} \left(1 + \frac{r}{m}\right)^m$$

$\searrow$

$$e^r$$



$$\left[\left(1 + \frac{r}{m}\right)^m \right]^t \rightarrow e^{rt}$$

$$r = 2\%$$

$$\left[\left(1 + \frac{r}{m}\right)^{\frac{1}{r/m}} \right]^r$$

$\uparrow$

$$= e^r$$

$$x = \frac{r}{m} \rightarrow 0$$

Derivatives

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$$y = f(\underline{x})$$

$$\begin{array}{ccc} \textcircled{x} & & \textcircled{y} \\ \Delta x \left(& & \right) \Delta y \\ & & \\ & & \underline{y + \Delta y} \\ & & \underline{\underline{\Delta y}} \end{array}$$

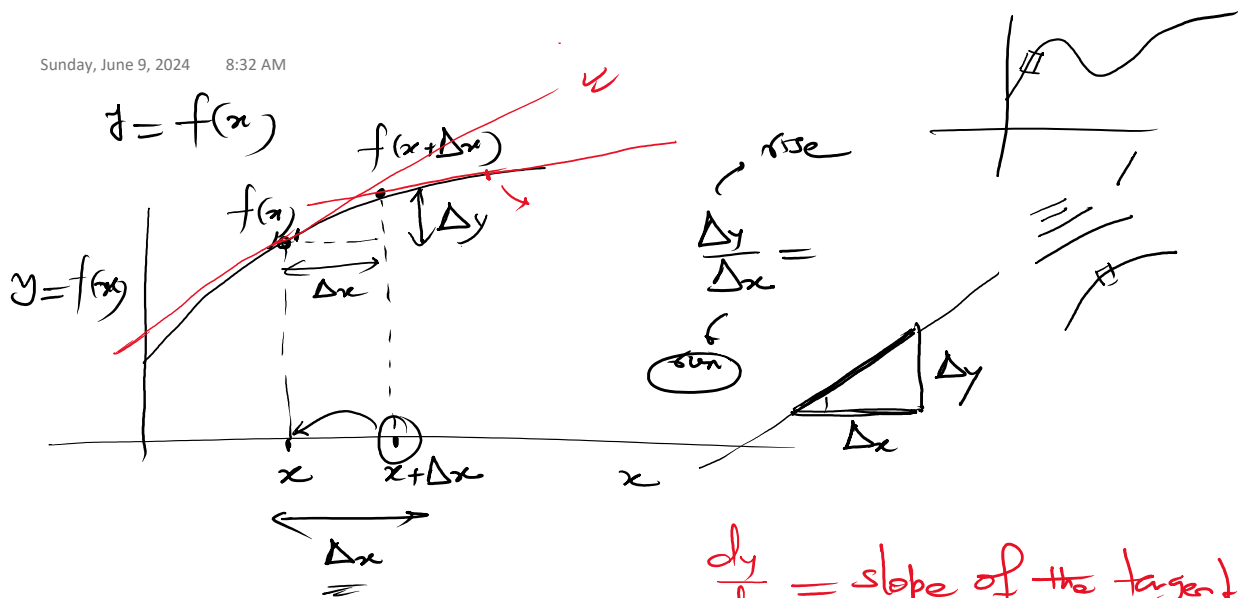
$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(\underline{x})}{\Delta x}$$

$$\rightarrow \boxed{\frac{dy}{dx}} = \boxed{\lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right)} \quad \left(\frac{0}{0} \right)$$

"derivative of y w.r.t. x"

Δ → difference

d → differential (infinitesimal change)

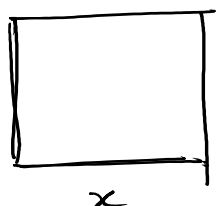


$$\left[\frac{dy}{dx} \right] = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right) \rightarrow \begin{array}{l} \text{rate of change of } y \\ \text{w.r.t } x \text{ / } \text{change in } x \\ \text{derivative of } y \\ \text{w.r.t } x. \end{array}$$

→ slope of the tangent to the curve

→ sensitivity of y w.r.t x (

Motivating example



$$A = x^2$$

$$\begin{array}{l} x \rightarrow x + \Delta x \\ A \rightarrow A + \Delta A \end{array}$$

$$\begin{aligned} \frac{\Delta A}{\Delta x} &= \frac{(x + \Delta x)^2 - x^2}{\Delta x} = \frac{(2x + \Delta x)(\cancel{\Delta x})}{\cancel{\Delta x}} \\ &= \underline{\underline{2x + \Delta x}} \end{aligned}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta A}{\Delta x} = \underline{\underline{2x}} = \frac{dA}{dx}, \quad \frac{d(x^2)}{dx} = 2x$$

$$\Delta A = \underline{\underline{(2x)(\Delta x)}} \quad \dots \text{valid if } \underline{\underline{\Delta x \rightarrow 0}}$$

first order approx.

$$\underline{\underline{\Delta A = (x + \Delta x)^2 - x^2}} \quad \text{exact}$$

$$y = a^x.$$

$$\ln(y) = x(\ln a)$$

$$\frac{d(\ln y)}{dx} = (\ln a)$$

$$\Rightarrow \frac{d(\ln y)}{dy} \cdot \frac{dy}{dx} = (\ln a)$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \ln a \quad \Rightarrow \frac{dy}{dx} = (a^x) \ln a.$$

$$y = x^{x^x} = x^y$$

$$\ln y = y \ln x$$

$$\frac{1}{y} y' = y \frac{1}{x} + \ln x \cdot y'$$

$$y' \left(\frac{1}{y} - \ln x \right) = \frac{y}{x}$$

$$\Rightarrow y' = \frac{y}{x \left(\frac{1}{y} - \ln x \right)}$$

$$y = x^x.$$

$$\ln y = x \ln x$$

$$\frac{1}{y} y' = x \cdot \frac{1}{x} + \ln x \cdot 1$$

$$y' = x^x (1 + \ln x)$$

Implicit derivative

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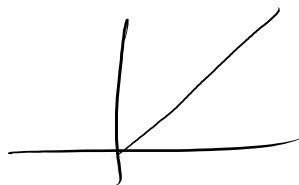
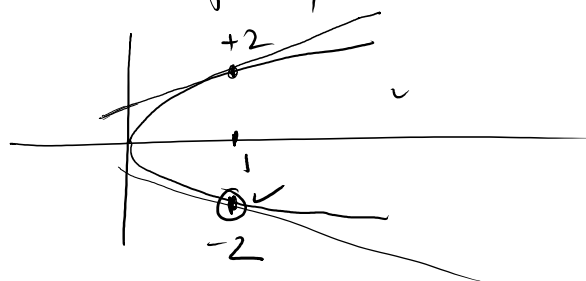
$$y = f(x) \quad \text{-- explicit}$$

$$\boxed{f(x, y) = 0}$$

$$\ln y + y^2 x - e^x = 0$$

$$\frac{1}{y} y' + y^2 \cdot 1 + \underline{x \cdot 2y y'} - e^x = 0$$

Find the slope of the tangent to the curve $y^2 = 4x$, at $(1, 2)$.

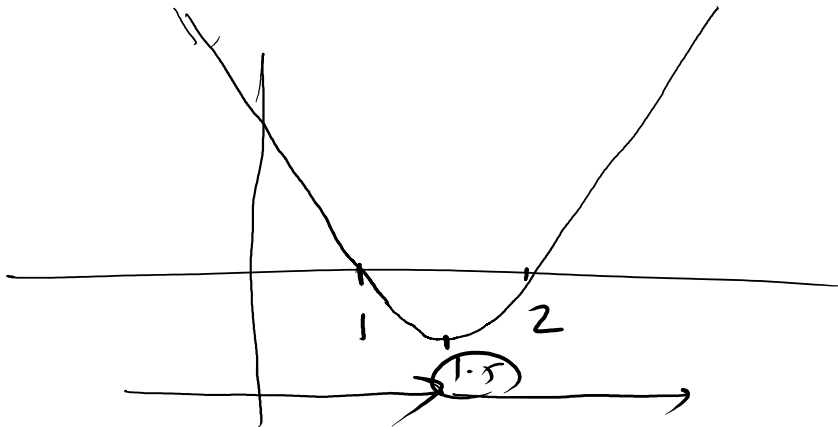


$$2yy' = 4$$

$$\Rightarrow y' = \frac{2}{y}$$

$$\left. \frac{dy}{dx} \right|_{(1,2)} = \frac{2}{2} = 1$$

$$\textcircled{2} \quad y = (x-1)(x-2) = \underline{\underline{x^2 - 3x + 2}}$$



$$x \rightarrow \infty, \quad y \rightarrow \infty \quad , \quad x \rightarrow -\infty, \quad y \rightarrow \infty$$

$$\frac{dy}{dx} = \underline{\underline{2x - 3}}$$

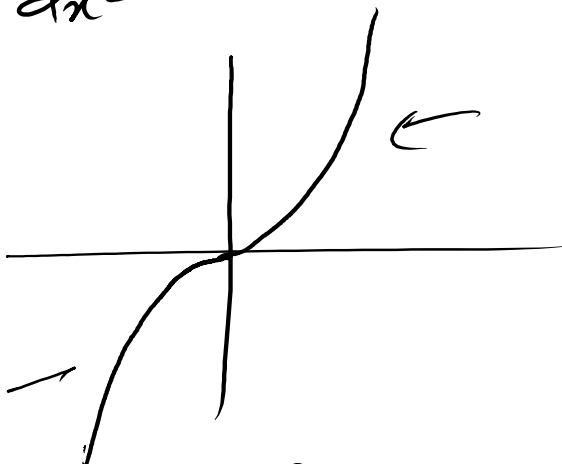
increasing if $2x - 3 > 0$
 $\Rightarrow \underline{\underline{x > 3/2}}$ $\underline{\underline{(3/2, \infty)}}$

decreasing if $2x - 3 < 0$
 $\Rightarrow \underline{\underline{x < 3/2}}$ $\underline{\underline{(-\infty, 3/2)}}$

$$\boxed{x = 3/2}$$

$$\frac{d^2y}{dx^2} = \underline{\underline{2}} > 0$$

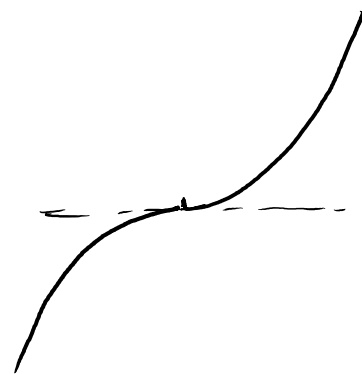
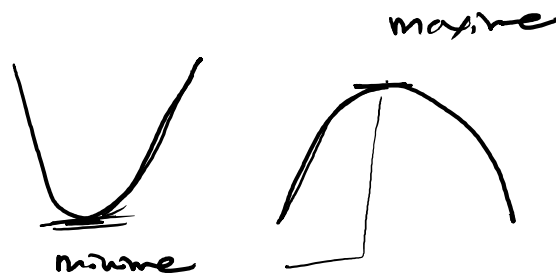
$$\frac{d^2y}{dx^2} = 0$$



$$f(x) = 3x^2 = 0,$$

$$6x|_{x=0} = 0$$

$$x=0$$



10^{101} , 101^{100} which is bigger = ?
 $\uparrow$

$$y = x^{1/x}$$

$$\ln y = \left(\frac{1}{x}\right) \ln x$$

$$\frac{1}{y} (y') = \left(\frac{1}{x}\right) \cdot \frac{1}{x} - \frac{(\ln x)}{x^2}$$

$$= \frac{1 - \ln x}{x^2}$$

increasing, $1 > \ln x$
 $\Rightarrow x < e$

decreasing if $x > e$

$$e = 2.71...$$

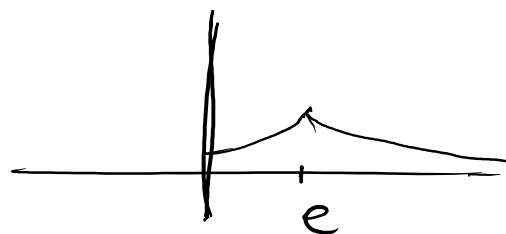
$$\boxed{e^{1/e}} > 3^{1/3}$$

$$10^{1/10} > \underline{101^{1/101}}$$

$101 \qquad 100$

$$\underline{4^7}, 7^4, 2^3, \underline{3^2}$$

$$\underline{11^{21}}, 21^{11}$$



$$(10^{1/10} \times 100)$$

$$1^{1/9} \dots 7^{1/7}$$

$$\Rightarrow 100^{101} > \underline{\underline{101^{100}}}$$

$$4^{\frac{1}{4}} > 7^{\frac{1}{7}}$$

$$\Rightarrow 4^7 > 7^4$$